

AI/ML intern DAY 11

TASK — 2D Virtual Try-On Research & Prototype

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Introduction

The objective of this task was to research and develop a basic AI-powered 2D Virtual Try-On prototype using Artificial Intelligence and Computer Vision technologies.

The system focuses on:

- clothing overlay on user images
- pose detection alignment
- body segmentation
- garment fitting logic
- image preprocessing

The AI system analyses human body posture and aligns clothing images on users using pose estimation and image processing workflows.

Clothing Overlay on User Images

Implemented basic garment overlay logic for placing virtual clothing on uploaded human body images.

The overlay system performs:

- garment positioning
- shoulder alignment
- body fitting
- clothing resizing

using AI body tracking.

Pose Detection Alignment

Used MediaPipe Pose Estimation for:

- shoulder tracking
- body posture detection

- skeletal landmark analysis
- garment alignment

The AI system detects:

- shoulder coordinates
- upper body position
- body orientation
- posture direction

to align clothing images correctly.

Body Segmentation

Researched body segmentation workflows for:

- separating body regions
- human parsing
- clothing region analysis
- overlay refinement

Segmentation helps improve:

- cloth fitting quality
 - body-cloth alignment
 - overlay accuracy
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Garment Fitting Logic

Studied garment fitting workflows including:

- dynamic cloth resizing
- shoulder-based alignment
- body contour matching
- overlay synchronization

The fitting logic adjusts clothing dimensions according to body landmark positions.

Image Preprocessing

Applied image preprocessing techniques including:

- image resizing
- RGB conversion
- background handling
- body image normalization

These preprocessing steps improve:

- pose estimation accuracy
- overlay quality
- segmentation consistency

Technologies Explored

Technology	Purpose
OpenCV	Image Processing
MediaPipe	Human Pose Detection
TensorFlow	AI Model Research
PyTorch	Deep Learning Research
Human Parsing Models	Body Segmentation

Workflow of the 2D Virtual Try-On System

Step 1 — Human Image Input

The AI system accepts uploaded human body images.

Step 2 — Pose Detection

MediaPipe detects:

- shoulders
- elbows
- hips
- body posture

using AI landmark tracking.

Step 3 — Body Segmentation

The AI system separates:

- body regions
- clothing regions
- background areas

for overlay processing.

Step 4 — Clothing Overlay

The garment image is resized and aligned using:

- shoulder coordinates
 - body position
 - body width
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Step 5 — Final Try-On Output

The AI system generates:

- final clothing overlay
 - skeletal visualization
 - aligned garment output
-

Basic 2D Virtual Try-On Python Code

Save file as:

virtual_tryon_2d.py

Python Implementation

```
import cv2
```

```
import mediapipe as mp
```

```
person = cv2.imread("person.jpg")
```

```
cloth = cv2.imread("shirt.png")
```

```
if person is None:
    print("person.jpg not found")
    exit()

if cloth is None:
    print("shirt.png not found")
    exit()

person = cv2.resize(person, (720,720))
cloth = cv2.resize(cloth, (250,250))

mp_pose = mp.solutions.pose
mp_draw = mp.solutions.drawing_utils

pose = mp_pose.Pose(
    static_image_mode=True,
    min_detection_confidence=0.7
)

rgb = cv2.cvtColor(person, cv2.COLOR_BGR2RGB)

results = pose.process(rgb)

if results.pose_landmarks:

    lm = results.pose_landmarks.landmark

    h, w, _ = person.shape

    ls = lm[11]
```

```
rs = lm[12]

x1, y1 = int(ls.x*w), int(ls.y*h)
x2, y2 = int(rs.x*w), int(rs.y*h)

center_x = int((x1+x2)/2)

cloth_width = abs(x2-x1) + 120

cloth = cv2.resize(
    cloth,
    (cloth_width, cloth_width)
)

y_offset = y1 - 40
x_offset = center_x - cloth.shape[1]//2

for i in range(cloth.shape[0]):
    for j in range(cloth.shape[1]):

        if (
            0 <= y_offset+i < person.shape[0]
            and
            0 <= x_offset+j < person.shape[1]
        ):

            person[
                y_offset+i,
                x_offset+j
            ] = cloth[i,j]
```

```

mp_draw.draw_landmarks(
    person,
    results.pose_landmarks,
    mp_pose.POSE_CONNECTIONS
)

print("Landmarks detected successfully")

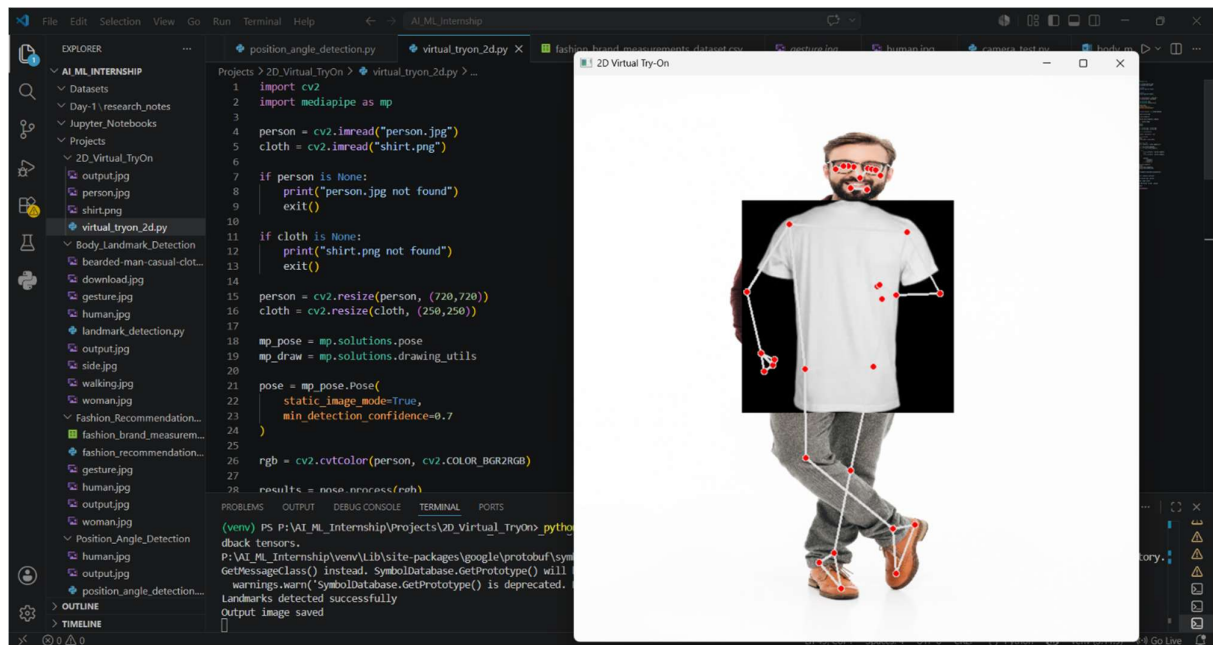
else:

    print("No landmarks detected")

cv2.imwrite("output.jpg", person)
print("Output image saved")

cv2.imshow("2D Virtual Try-On", person)
cv2.waitKey(0)
cv2.destroyAllWindows()

```



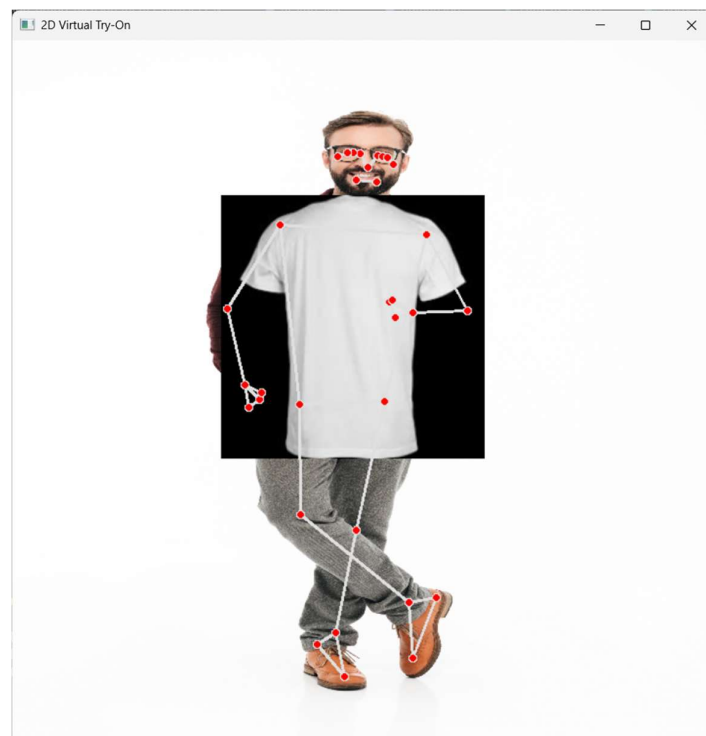
Expected Output

The AI system generates:

- clothing overlay output
- human landmark visualization
- aligned garment fitting
- skeletal tracking
- virtual try-on image

The final output image is automatically saved as:

output.jpg



Research Findings

The research helped understand:

- AI clothing alignment
- body segmentation workflows
- garment fitting systems
- pose estimation technologies
- image preprocessing techniques

The system demonstrated how Artificial Intelligence can generate virtual clothing fitting outputs using body landmark analysis and Computer Vision workflows.

Learning Outcomes

During this task, the following concepts and skills were learned:

- Human pose estimation
- Clothing overlay systems
- Body segmentation workflows
- AI garment fitting
- Skeletal landmark tracking
- Virtual try-on technologies
- Computer vision applications

Conclusion

The 2D Virtual Try-On Research and Prototype task successfully demonstrated how Artificial Intelligence and Computer Vision technologies can generate basic virtual fitting systems using body pose estimation and garment overlay workflows.

The task improved understanding of:

- clothing overlay systems
- pose detection alignment
- body segmentation
- garment fitting logic
- image preprocessing
- AI fashion technologies
- virtual fitting workflows